Application: U-Joint Monitoring

The Challenge

A U-Joint manufacturer was having failures with U-Joints installed on truck fleets. They wanted to determine if the cause was due to environmental conditions or road conditions, and at what point the units were failing. To test this, they wanted to obtain accurate measurements of both torque and temperature on units in the field.

System Requirements

There were several requirements for this monitoring system that drove the company to a TECAT solution.

- ✓ Compact: The systems had to be mounted on U-Joints, with a very small footprint.
- ✓ Portable: Installation and measurements had to be done in the field, so the measurement systems needed to be easily transported.
- ✓ Power Efficient: This application involved long term monitoring in a tight space with difficult access. The system needed to be able to run for long periods of time without recharging the battery.

The TECAT Solution

TECAT installed the WISER torque measurement system with the optional temperature sensor on new U-Joints, which were then shipped to the trucks in the field for installation. The long term monitoring did not require high-speed data acquisition, so the innately low power consumption of WISER was further improved by reducing data rates. By monitoring both the strain gauge based torque measurements and the temperature simultaneously, the company was able to determine the cause of the failures, and engineer the proper solution.

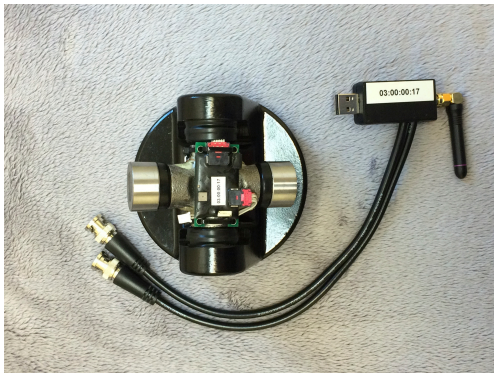


Figure 1 WISER Installation on a U-Joint